Semisupervised Cross-Domain Remote Sensing Scene Classification via Category-Level Feature Alignment Network

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***Abstract*— In recent years, remote sensing scene classification has obtained much attention owing to its widespread applications. Nevertheless, the existing deep learning-based scene classification methods suffer from domain shift. Given a few labeled target domain samples, semisupervised domain adaptation (SSDA) has been explored to enhance the generalization ability of the neural network across domains. However, current SSDA methods focus on domain alignment but fail to achieve fine-grained category-level feature alignment. In this article, we propose a category-level feature alignment network (CFAN), which uses a dual-directional prototype alignment module to achieve category-level feature alignment of the source and target domains. In particular, we utilize pseudo-labeling technique to generate pseudo labels for unlabeled target domain features and build target domain prototypes. Nonetheless, the model performance suffers from the noise conveyed in the pseudo labels. To tackle this issue, we further employ the adversarial training strategy to align the multimodal information of different domains and thus improve the quality of pseudo labels. The proposed CFAN outperforms state-of-the-art (SOTA) methods on commonly used remote sensing datasets, with respective mean accuracies of 92.9%, 93.3%, and 94.3% with EfficientNet_B0, VGG16, and ResNet34 backbones in12 adaptation scenarios.**

***Index Terms*— Adversarial training, scene classification, semisupervised domain adaptation (SSDA).**

I. Introduction

REMOTE sensing image scene classification has applications in areas such as urban planning [1], disaster monitoring [2], and land use [3]. Deep learning-based methods have been proposed for remote sensing scene classification, including those based on transformers [4], generative adversarial networks [5], [6], convolutional neural networks [7], and autoencoders [8]. Such methods assume that training and testing samples are independently drawn from the same distribution. However, in practice, these are often collected from quite different distributions, with variations in capturing angle, optical sensors, and geographic regions, where a trained deep neural network tends to show a degraded performance, which is referred to as domain shift.

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To address the above issue, unsupervised domain adaptation (UDA) [9], [10] and semisupervised domain adaptation (SSDA) methods [11], [12], [13] have been proposed to bridge the domain shift between the source and target domain. In UDA, the network is trained using the labeled source samples and unlabeled target samples to reduce the domain shift. However, the generalization ability of UDA is limited due to the lack of labeled target samples. Compared with UDA, the neural network is trained on rich-annotated source samples and a few labeled target samples in SSDA. Hence, SSDA is a feasible solution to improve the generalization ability of the neural network across domains.

The existing SSDA methods [14], [15], [16] in remote sensing field explore moment matching and adversarial training to perform feature alignment and thus to boost the classification performance in the target domain. Moment matching methods [14] align the distributions of source and target domains by the hand-crafted statistics, such as maximum mean discrepancy (MMD) [17] and CORAL [18]. However, the hand-crafted statistics are applied on the extracted features and rely on the hypothesis of the feature distributions, neglecting the domain information. Adversarial training-based methods [15], [16] introduce a discriminator to distinguish the domain label of extracted features to fully explore the domain information. The feature extractor learns domain-invariant features to confuse the trainable discriminator. As a result, adversarial training can achieve a better domain alignment than moment matching. Despite the significant improvement, moment matching and adversarial training only perform global feature alignment. As complex multimodal structures naturally exist in the distribution of remote sensing images, the above methods possibly fail to capture such multimodal structures to achieve discriminative alignment. Even if the discriminator is fully confused or the moments are matched seamlessly, there is no guarantee that two distributions are similar enough in category level. On the other hand, category-level feature alignment is more fine-grained and robust than domain alignment. Nonetheless, the above methods ignore the category-level information and exhibit the limited performance.

To address the above issues, we propose a semisupervised cross-domain remote sensing scene classification method using a category-level feature alignment network (CFAN). We first design a dual-directional prototype alignment module to achieve fine-grained category-level feature alignment and reduce category differences between the source and

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target domains. We extract features from the source and target images, and store these in the respective source and target memory banks, which are used to build the source and target category prototypes. Pseudo-labeling technology is used to label the unlabeled target domain features. The trained model makes predictions, and generate a one-hot label for the unlabeled target features, which helps to build target domain prototypes. Furthermore, to reduce noisy pseudo labels and build more accurate target domain prototypes, we introduce the adversarial training module to acquire domain-invariant features and align the multimodal structures of different domains. We take the L_2 distance between target domain prototypes and target samples to measure the quality of target domain prototypes. The boxplots show that the quality of target domain prototypes is significantly improved when adversarial training is applied. With the pseudo-labeling technique and adversarial training module, CFAN outperforms the previous SSDA method for remote sensing images, with a mean accuracy of 94.3% with ResNet34 over 12 adaptation scenarios.

The main contributions of this article can be summarized as follows.

1) To achieve fine-grained category-level alignment, a dual-directional prototype alignment module is proposed to align the acquired features from the source samples to the target category prototypes, and from the target samples to the source category prototypes.
2) To build accurate target prototypes, we introduce a technique to generate pseudo labels for target samples, and employ adversarial training to extract domain-invariant features to improve their quality.
3) Extensive experiments show that the proposed method advances the state-of-the-art.

The rest of this article is organized as follows. Section II reviews related work. Section III describes the proposed method. We describe our experiments and discuss their results in Section IV. We present our conclusions in Section V.

II. Related Work

A. Semisupervised Image Classification

Semisupervised learning (SSL) focuses on alleviating the need for large amounts of labeled data, which greatly reduces manual annotation. Numerous semisupervised methods for image classification have recently been proposed. Based on the smooth assumption [19], most concentrate on consistency regularization, applying different transformations to input images, and then leveraging the semantic label consistency.

In early work, temporal ensemble (TE) [20] used the exponential moving average (EMA) of a model to produce pseudo labels. However, TE was time-consuming when trained with large-scale datasets. To promote efficiency, Tarvainen and Valpola [21] averaged the model weights instead of label predictions. Recently, some work has adopted data from strong-augmentation methods, such as UDA [22], RandAugment [23], CTAugment [24], and Cutout [25], to build the consistency loss between weak- and strong-augmented images. These strong-augmentation methods have been proven effective [26], [27]. Sohn et al. [26] proposed FixMatch and adopted RandAugment and CTAugment as strong augmentation, and standard flip-and-shift as weak augmentation. The model was expected to provide consistent predictions even with different input perturbations. More augmentations were needed in some circumstances to provide more diversity of images. Gui et al. [28] selected a naive sample and applied more diverse augmentation to make it more effective for model training. However, image-level augmentation was limited by the manual design, which was inadequate to explore wider augmentation space. Yang et al. [29] proposed an auxiliary feature perturbation stream to expand the perturbation space, and a dual-stream perturbation technique to increase diversity. Although semisupervised methods have achieved success in image classification task, they usually show deteriorated performance when the labeled images and the unlabeled images are drawn from different distributions.

B. Adversarial Training

Adversarial training is a machine learning technique that is widely used to improve the robustness and generalization of models. A model is trained with a combination of adversarial examples, which are input data that have been intentionally perturbed to cause errors. The model learns to better distinguish between real-world samples and subtly manipulated adversarial examples. Goodfellow et al. [30] first introduced a fast gradient sign method (FGSM) to generate adversarial examples effectively. To improve the quality of the generated adversarial examples, Jia et al. [31] and Kurakin et al. [32] proposed prior-guided FGSM and PGD, respectively.

In the field of remote sensing, adversarial examples were used to improve the robustness of the model. Czaja et al. [33] first introduced adversarial training to remote sensing. They added a patch of noisy pixels to satellite images, and demonstrated that adversarial examples reduced the robustness of the model. Nevertheless, Czaja et al. [33] could not determine the distribution of adversarial examples. Cheng et al. [34] proposed perturbation-seeking generative adversarial networks (PSGANs). PSGANs modeled the distribution of perturbations in adversarial examples to improve the defense ability of the model. Different from PSGANs, Shi et al. [35] and Tu et al. [36] improved the robustness of the model from feature aspect. More reliable feature was selected to efficiently defend against adversarial examples. However, these methods utilize adversarial training to improve the generalization performance of deep learning model, instead of obtaining domain-invariant features in different domains explicitly.

C. Semisupervised Domain Adaptation

Domain adaptation (DA) has drawn much attention recently. It aims to bridge the discrepancy between the source and target domains by acquiring domain-invariant features. Although UDA is attractive, the generalization performance of UDA is limited due to the lack of labeled target domain samples. SSDA is proposed to improve the generalization performance by labeling a few target domain samples. The existing SSDA methods learn domain-invariant features by moment matching or adversarial training. Moment matching methods align

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different distributions via some hand-crafted statistics, such as MMD [17], Kullback–Leibler (KL) divergence [37], and CORAL. Specifically, Huang et al. [14] aligned the labeled source and target samples with unlabeled target samples with MMD. Inspired by GAN, adversarial training was introduced to extract domain-invariant feature for domain alignment. Wang et al. [15] extended ADDA from UDA to SSDA. Lasloum et al. [38] applied MME [11] to a multi-source SSDA task. To further extract discriminative features for the classification of target samples, Teng et al. [16] used two different classifiers as discriminator that makes predictions for target samples. However, the above methods only focus on domain alignment, category differences of source and target domains are ignored.

III. Methodology

In this section, we define SSDA and introduce our method in detail. The workflow of overall framework is illustrated in Fig. 1.

A. Problem Definition

In SSDA, we are given labeled samples collected from the source and target domains, denoted as $D_s = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and $D_t^l = \{(x_i^l, y_i^l)\}_{i=1}^{N_l}$, respectively, and unlabeled samples $D_t^u = \{(x_i^u)\}_{i=1}^{N_u}$ collected from the target domain. $x \in \mathbb{R}^{H\times W\times 3}$ is an image with semantic label y, and N is the number of images. In the target domain, the number of unlabeled images N_u is much larger than the number of labeled samples N_l. We assume that the source and target domains share the same label space, but have different distributions in image space. SSDA methods aim to improve generalization ability with a lower annotation cost than traditional UDA.

B. Basic Supervision

Inspired by the success of consistency regularization in SSL [26], [27], [39], [40], consistency regularization has been introduced into the SSDA. We follow the recent SSDA methods [14], [41] and use two image augmentation methods to promote the diversity of images. The model is encouraged to make the same predictions for the augmented images, which improves the generalization ability of the model. RandAugment [23] is used for strong augmentation, and randomly cropping and flipping for weak augmentation. Images sampled from the source domain are only weakly augmented, while target images are processed using both weak and strong augmentations. Following the common practice in SSDA methods [42], [43], a neural network is utilized to supervise the labeled images.

Weak-augmented source/target images and strong-augmented target images are sent into the feature extractor $\mathcal{F}$ to generate the corresponding features. $\mathbf{f}_w^s$, $\mathbf{f}_w^l$, and $\mathbf{f}_w^u$ are sent into the classifier $\mathcal{G}$ followed by the softmax function, to obtain the corresponding class predictions $\mathbf{g}_w^s$, $\mathbf{g}_w^l$, and $\mathbf{g}_w^u$.

The cross-entropy loss $\mathcal{L}_{\text{ce}}$ is described as follows:

$$\mathcal{L}_{\text{ce}} = -\frac{1}{N_s+N_l}\sum_{i=1}^{N_s+N_l}\sum_{c=1}^{C} y_{c,i}^{s,l}\log\left(\mathbf{g}_{w,c,i}^{s,l}\right) \tag{1}$$

where $\mathbf{g}_{w,c,i}^{s,l}$ is the prediction of the weak-augmented source and labeled target images in the cth category.

C. Dual-Directional Prototype Alignment Module

Basic supervision minimizes the empirical risk of labeled source and target samples. According to the DA theory [44], we further extract robust domain-invariant features. Following BSCA [14], we build a dual-directional prototype alignment module via the memory bank mechanism, as illustrated in Fig. 2, consisting of a source memory bank $\mathcal{B}_s = \{B_s^c\}_{c=1}^C$ and a target memory bank $\mathcal{B}_t = \{B_l^c, B_u^c\}_{c=1}^C$, where c is the index of the category label, and B_s^c, B_l^c, and B_u^c are the memory queues of source domain images, labeled target domain images, and unlabeled target domain images, respectively. Taking $\mathcal{B}_s$ as an example, we build a dictionary for the memory bank, where keys denote categories, and B_s^c is the value corresponding to the memory queue with length L for the cth category. The source domain category prototypes are calculated by averaging the features in the memory queues. The target memory bank $\mathcal{B}_t$ follows the same process. Note that B_l^c and B_u^c make up the target memory bank $\mathcal{B}_t$. B_l^c plays a key role in generating robust target prototypes, which provide reliable features of the corresponding categories. The length of B_l^c is $2 \times N_l$, and thus the length of B_u^c is $L - 2 \times N_l$. The memory bank mechanism includes three operations: construction, updating, and alignment, which we discuss below.

1) Construction: The initial states of the source memory bank $\mathcal{B}_s$ and target memory bank $\mathcal{B}_t$ are set to zero. Then the source domain images x^s, labeled target domain images x^l, and unlabeled target domain images x^u are fed into the feature extractor $\mathcal{F}$, which generates the corresponding features. We store $\mathbf{f}_w^s$ in the source memory queue B_s^c, and $\{\mathbf{f}_w^l, \mathbf{f}_w^u\}$ in the target memory queue B_t^c. Consequently, the category prototypes p_s^c and p_t^c are calculated by averaging all the elements of the category in the corresponding memory queues B_s^c and $\{B_l^c, B_u^c\}$. In this way, the limited samples in each minibatch are accumulated to build accurate and robust category prototypes

$$\begin{cases} p_s^c = \dfrac{1}{L}\displaystyle\sum_{i=1}^{L} B_s^{ci} \\ p_t^c = \dfrac{1}{L}\left(\displaystyle\sum_{i=1}^{2\times N_l} B_l^{ci} + \sum_{i=2\times N_l}^{L} B_u^{ci}\right). \end{cases} \tag{2}$$

2) Update: To improve the robustness of the categorywise prototypes for alignment, we update the memory bank continuously in first-in-first-out (FIFO) order. We append the new extracted features in the corresponding memory queue, and discard the oldest features after each iteration if its length reaches a specified length L. B_s^c and B_l^c are updated by the corresponding features $\mathbf{f}_w^s$ and $\mathbf{f}_w^l$. However, we update the unlabeled target domain memory queue B_u^c with $\mathbf{f}_w^u$, which is pseudo labeled (Section III-D). Note that there are no substantial features fed to the memory banks, because several elements remain zero at the early training stage. In this case, we calculate the prototypes only with the nonzero elements.

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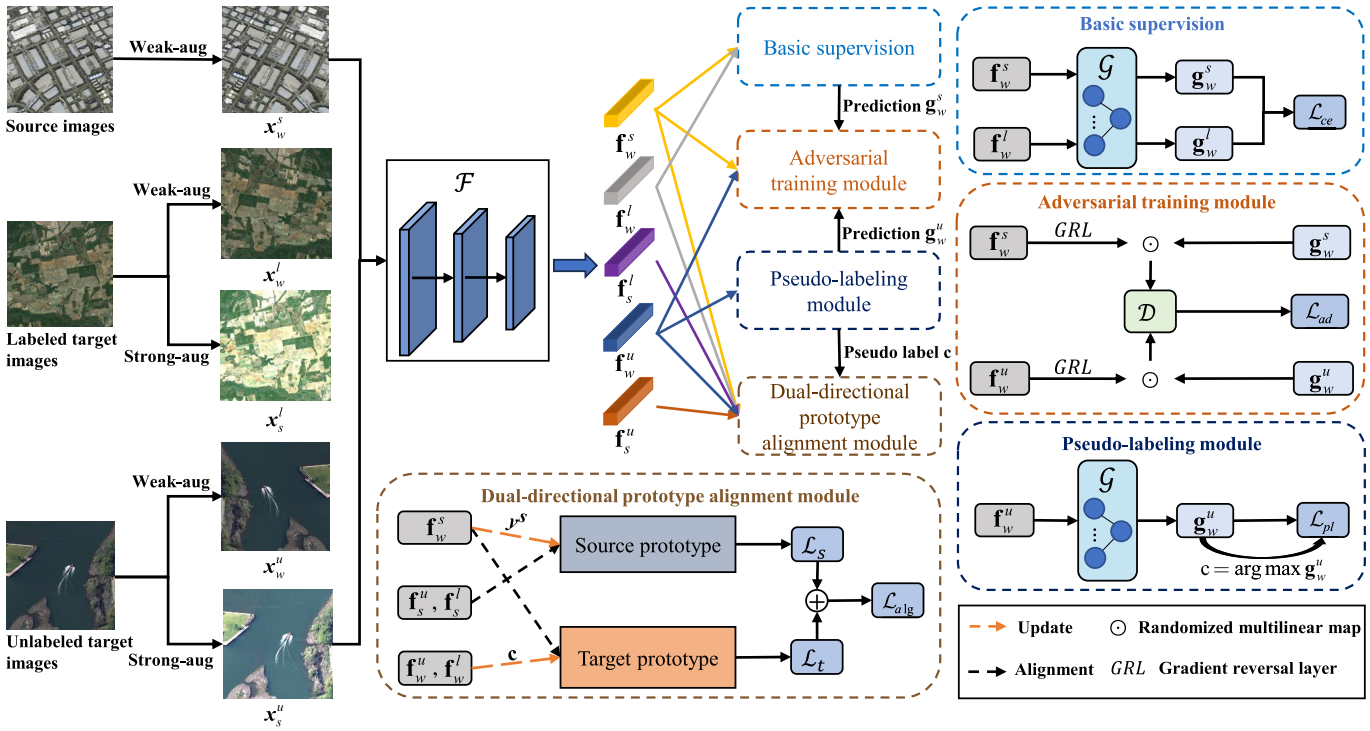

Fig. 1. Workflow of overall framework. Images are augmented before being sent into the network. Feature extractor $\mathcal{F}$ and classifier $\mathcal{G}$ generate features $\mathbf{f}$ and class predictions $\mathbf{g}$, respectively. In basic supervision, predictions of labeled source and target samples are used to build cross-entropy loss $\mathcal{L}_{\text{ce}}$. There are three modules. (1) Dual-directional prototype alignment module achieves category-level feature alignment by aligning features of source/target samples with target/source prototypes. (2) Pseudo-labeling module generates pseudo labels for weak-augmented unlabeled target samples. Pseudo labels are used to build target prototypes. (3) Adversarial training module aligns multimodal information of different domains with the help of predictions $\mathbf{g}_w^u$ and $\mathbf{g}_w^s$, helping to improve the quality of pseudo labels.

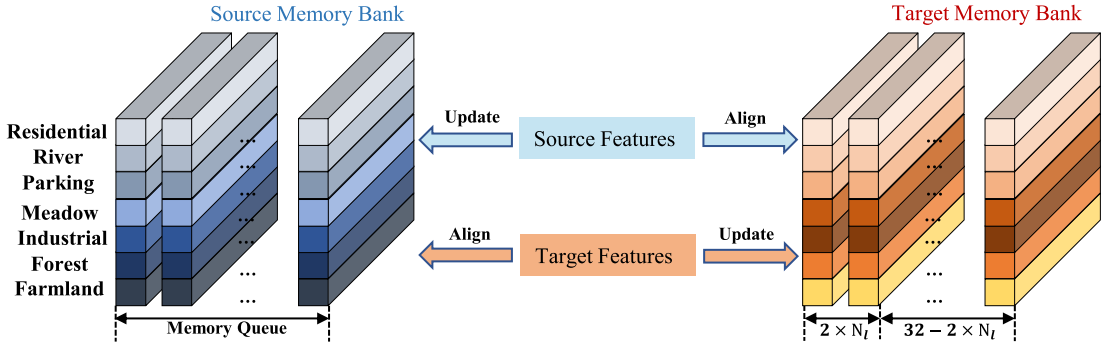

Fig. 2. Memory bank mechanism.

3) Alignment: The proposed memory bank mechanism is designed to align the feature distributions of the source and target domains. To this end, we design the dual-directional prototype alignment loss $\mathcal{L}_{\text{alg}}$, which has two parts. First, the features of the source images $\mathbf{f}_w^s$ are aligned with the target domain prototypes using the loss $\mathcal{L}_s$

$$\mathcal{L}_s = \frac{1}{N_s}\sum_{i=1}^{N_s}\left(\mathbf{f}_{w,i}^s - p_t^c\right)^2. \tag{3}$$

The mean square error (MSE) is used to compute the loss between the feature and prototype. Then the features of target images, including $\mathbf{f}_s^l$ and $\mathbf{f}_s^u$, are aligned with the source domain prototypes using the loss $\mathcal{L}_t$

$$\mathcal{L}_t = \frac{1}{N_l + M_u}\left[\sum_{i=1}^{N_l}\left(\mathbf{f}_{s,i}^l - p_s^c\right)^2 + \sum_{i=1}^{M_u}\left(\mathbf{f}_{s,i}^u - p_s^c\right)^2\right] \tag{4}$$

where M_u is the number of samples whose $\mathcal{E} < \tau$, and c is the index of the category label/pseudo label corresponding to the feature $\mathbf{f}_{s,i}^l/\mathbf{f}_{s,i}^u$.

In BSCA, features of labeled target samples are aligned with target prototypes, while only features of unlabeled target samples are aligned with source domain prototypes. The lack of labeled target samples leads to the loss of category information to some extent, which obstructs accurate and comprehensive alignment in feature space. Different from BSCA, we concatenate $\mathbf{f}_s^l$ and $\mathbf{f}_s^u$, and align both labeled and unlabeled target domain features with the source domain prototypes, to fully exploit target domain samples. In this way, we maintain all the category information of target domain, which enables more robust category-level feature alignment. The overall alignment loss $\mathcal{L}_{\text{alg}}$ is given as follows:

$$\mathcal{L}_{\text{alg}} = \mathcal{L}_s + \mathcal{L}_t. \tag{5}$$

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D. Pseudo-Labeling Module

Since the number of labeled target domain samples is small, it is difficult to build representative target domain prototypes during training. To address this issue, a pseudo-labeling module is used to label the target domain samples and retrain the model with high-confidence pseudo-labeled samples. Specifically, the maximum-predicted category is utilized as the pseudo label of unlabeled target domain samples

$$c = \arg\max \mathbf{g}^u_{w,i} \tag{6}$$

where $\mathbf{g}^u_{w,i}$ is the class distribution of unlabeled target images.

We compute the entropy [45] of unlabeled target images as follows:

$$\mathcal{E} = -\sum_{i=1}^{C} \mathbf{g}^u_{w,i} log \mathbf{g}^u_{w,i}. \tag{7}$$

Then the pseudo-label loss is given as follows:

$$\mathcal{L}_{\text{pl}} = -\sum_{\mathbf{x}^u \in D^u_t} \mathbb{1}(\mathcal{E} < \tau) \cdot c \cdot \log \mathbf{g}^u_{w,c} \tag{8}$$

where $\mathbb{1}(\cdot)$ is an indicator function. We adopt the entropy $\mathcal{E}$ to select high-quality pseudo-labeled samples.

The pseudo-label loss is utilized to increase the number of labeled target samples and build representative target domain prototypes, which enables better alignment between source domain samples and target domain prototypes. Also, the generalization ability of the model is improved to some extent with the high-confidence pseudo-labeled target domain samples.

E. Adversarial Training Module

Although the pseudo-labeling module increases the number of target domain samples for the construction of target domain prototypes, the quality of the generated pseudo labels is unsatisfactory. BSCA suffers from low-quality pseudo labels, which limits the performance of feature alignment. Different from BSCA, we use adversarial training to improve the quality of pseudo labels and generate accurate target prototypes for alignment. Adversarial training [46] is an effective way to acquire robust, domain-invariant features and align the multimodal structures of different domains.

In adversarial training module, we introduce a domain discriminator $\mathcal{D}$ that receives image features and tries to distinguish source and target samples, while the feature extractor $\mathcal{F}$ is trained to extract features $\mathbf{f} = \mathcal{F}(x)$ that confuse the domain discriminator. Finally, the classifier $\mathcal{G}$ receives image features and predicts class distributions, i.e., $\mathbf{g} = \sigma(\mathcal{G}(\mathbf{f}))$, where $\sigma(\cdot)$ is the softmax function. The two-player game can be described as follows:

$$\min_{\mathcal{F},\mathcal{G}} \max_{\mathcal{D}} \mathcal{L}_{\text{ce}} + \lambda \mathcal{L}_{\text{ad}}\left(\mathcal{D}(\mathbf{f}^{s,u}_w), y^{s,u}\right) \tag{9}$$

where $\mathbf{f}^{s,u}_w = \mathcal{F}(x^{s,u}_w)$ indicates features of weak-augmented source and unlabeled target images, and λ is a trade-off between adversarial loss and labeled image classification loss.

In the process of adversarial training, the adversarial loss is given as follows:

$$\mathcal{L}_{\text{ad}} = -\frac{1}{N_s}\sum_{i=1}^{N_s} \log \mathcal{D}\left(\mathbf{f}^s_{w,i}\right) - \frac{1}{N_u}\sum_{j=1}^{N_u} \log\left(1 - \mathcal{D}\left(\mathbf{f}^u_{w,j}\right)\right). \tag{10}$$

The gradient reversal layer (GRL) is adopted to realize the $\max(\cdot)$ operation.

We acquire domain-invariant features from (10), which focuses on aligning the global feature distributions. However, only adapting the features extracted by $\mathcal{F}$ is limited if the joint distributions of the feature and class are not identical in the source and target domains, since the extracted features become increasingly specific with the increase in network depth. Even if the discriminator is successfully deceived, it is still impossible to ensure that the feature distributions of the source and target domains are aligned, due to the multimodal nature of images.

Motivated by conditional generative adversarial networks (CGANs) [5], Long et al. [46] found that the classifier prediction $\mathbf{g}$ conveys discriminative information that can be used to assist in DA. Following Long et al. [46], we condition the discriminator on the cross covariance of feature representation $\mathbf{f}$ and classifier prediction $\mathbf{g}$, extending the two-player game in (9) as follows:

$$\min_{\mathcal{F},\mathcal{G}} \max_{\mathcal{D}} \mathcal{L}_{\text{ce}} + \lambda \mathcal{L}_{\text{ad}}(\mathcal{D}(T(\mathbf{h}^{s,u})), y^{s,u}) \tag{11}$$

where $T(\mathbf{h}^{s,u})$ is a multilinear map function, which can capture the multimodal information of different distributions. Furthermore, to avoid dimension explosion in the multilinear map function, a randomized method [47] is applied in the multilinear map

$$T(\mathbf{h}) = \begin{cases} T_{\otimes}(\mathbf{f}, \mathbf{g}), & \text{if } d_{\mathbf{f}} \times d_{\mathbf{g}} \leq 4096 \\ T_{\odot}(\mathbf{f}, \mathbf{g}), & \text{else} \end{cases} \tag{12}$$

where $d_{\mathbf{f}}$ and $d_{\mathbf{g}}$ are the respective dimensions of features and predictions. When $d_{\mathbf{f}} \times d_{\mathbf{g}} \leq 4096$, we select multilinear map $T_{\otimes}$, and otherwise, the randomized multilinear map $T_{\odot}$.

Adversarial training extracts domain-invariant features and aligns the multimodal structures of different domains, which reduces the noise of pseudo labels. This further improves the quality of target domain prototypes. This is analyzed in Section IV-E.

F. Overall Loss

The overall training loss of CFAN includes the basic supervision loss $\mathcal{L}_{\text{ce}}$, the alignment loss $\mathcal{L}_{\text{alg}}$, the pseudo-label loss $\mathcal{L}_{\text{pl}}$, and the adversarial loss $\mathcal{L}_{\text{ad}}$

$$\mathcal{L} = \mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{pl}} + \alpha \mathcal{L}_{\text{alg}} + \lambda \mathcal{L}_{\text{ad}} \tag{13}$$

where α and λ are set to 0.02 [14] and 1 [46], respectively.

IV. Experiments

We introduce the datasets used in our experiments, and discuss their details. Finally, the comparison experiments and ablation study are discussed.

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Fig. 3. Seven common classes from WHU, RSSCN7, NWPU-RESISC45, and AID.

A. Datasets

NWPU-RESISC45 [48] (*N*) is introduced by the Northwestern Polytechnical University in 2017. The dataset was extracted from Google Earth imagery, the 45 classes of which contain an average of 31 500 images each. This challenging remote sensing scene classification dataset has varying spatial resolutions, complicated backgrounds, and viewpoint variations.

AID [49] (*A*) is a large-scale dataset, comprising 10 000 images from 30 classes. It is proposed by the Wuhan University in 2016. Images are 600 × 600, and were acquired from various countries and regions.

RSSCN7 [50] (*R*) contains 2800 images acquired from different weathers and seasons, with scales of 1:700, 1:1300, 1:2600, and 1:5200. The seven classes are farmland, forest, industrial, meadow, parking, river, and residential, each with 400 images of size 400 × 400.

WHU-RS19 [51] (*W*) is composed of satellite images exported from Google Earth, which can produce high-resolution satellite images as sharp as 0.5 m. It contains 19 categories of scenes, each with 50 images.

Table I shows the number of samples from seven common classes in each dataset, and Fig. 3 shows the samples. These classes are farmland, forest, industrial, meadow, parking, river, and residential. We built 12 adaptation scenarios, as all datasets can be used as the source and target domain. In the target domain, N_l images per class, with their labels, were selected as labeled samples, and six samples per class were used as a validation set.

TABLE I
NUMBERS OF SAMPLES OF SEVEN COMMON CLASSES IN DATASETS

Class	RSSCN7	AID	NWPU-RESISC45	WHU-RS19
Residential	400	410	700	54
River	400	410	700	56
Parking	400	390	700	50
Meadow	400	280	700	61
Industrial	400	390	700	53
Forest	400	250	700	53
Farmland	400	370	700	50
Resolution (m)	-	0.5–8	0.2–30	0.5
Size	400 × 400	600 × 600	256 × 256	600 × 600
Source	Google Earth	Google Earth	Google Earth	Google Earth
Year	2015	2016	2017	2010

B. Implementation

We conducted experiments on three commonly used backbones, including VGG16 [57], ResNet34 [58], and EfficientNet_B0 [59], pretrained on ImageNet [60]. Images were resized to 256 × 256, and randomly cropped to 224 × 224. SGD was selected as our optimizer, with momentum and weight decay set to 0.9 and 0.0005, respectively. The learning rate was 0.001, and the batch size was 24. We set the length L of the memory queue to 32, and the labeled target memory queue was $2 \times N_l$. Following the common practice, we set N_l to 3. We set the entropy threshold τ in (8) to 0.5, and trained the model for 50 000 iterations in all adaptation scenarios. All experiments were conducted on a server deployed with an Nvidia RTX 2080Ti GPU.

C. Comparison Experiments

We evaluated the proposed CFAN in 12 remote sensing adaptation scenarios, comparing it with baselines including the SSL methods, UDA methods, and SSDA methods.

1) Baselines: **Fixmatch** [26] is a fundamental method in SSL, which achieved the state-of-the-art (SOTA) in semisupervised classification of natural images. **ECB-FAM** [53]

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TABLE II
RESULTS OF CFAN WITH VGG16 AS BACKBONE UNDER THREE-SHOT SETTING

Type	Method	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean
SSL	FixMatch [52]		77.9			96.3			67.5			93.4		83.8
UDA	ECB-FAM [53]	90.5	74.6	80.6	93.9	79.5	92.3	69.7	71.0	68.2	94.1	97.7	87.6	83.3
SSDA	S+T	92.6	83.6	88.0	94.1	88.9	93.8	72.9	72.4	76.9	96.9	98.3	92.4	87.6
	ADDA [54]	81.8	69.2	73.9	90.5	82.4	77.9	61.8	64.8	66.7	86.1	89.6	80.9	77.1
	RevGrad [38]	88.2	81.7	90.0	93.9	88.9	89.8	73.8	71.9	77.5	96.9	98.3	93.4	87.0
	SSDAN [38]	94.6	84.7	89.5	96.8	94.2	95.5	79.6	80.6	81.8	98.3	99.3	96.9	91.0
	ENT [55]	86.5	72.0	86.9	92.2	86.0	91.7	60.4	68.0	73.0	80.2	98.6	89.9	82.1
	CDAC [56]	85.6	74.6	71.7	89.9	81.8	79.5	65.0	70.2	69.8	81.2	89.2	73.6	77.7
	BSCA [14]	**96.5**	**94.2**	**95.9**	96.5	**94.5**	96.5	78.0	**82.8**	84.7	98.6	99.0	**99.3**	93.0
	ours	95.8	92.0	95.8	**96.7**	93.6	**96.8**	**84.0**	81.2	**85.9**	**99.3**	**99.0**	99.0	**93.3**

TABLE III
RESULTS OF CFAN WITH RESNET34 AS BACKBONE UNDER THREE-SHOT SETTING

Type	Method	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean
SSL	FixMatch [52]		81.3			95.0			63.5			97.6		84.4
UDA	ECB-FAM [53]	88.5	74.5	83.2	92.5	77.5	92.8	66.9	68.9	65.4	91.7	97.6	79.9	81.6
SSDA	S+T	89.9	76.4	84.4	91.7	84.5	91.6	71.4	68.4	69.4	92.0	97.6	92.4	87.6
	ADDA [54]	87.3	78.6	83.1	77.2	76.5	90.7	64.7	72.4	65.0	89.9	94.8	85.8	80.5
	RevGrad [38]	90.2	80.2	86.9	93.5	86.8	91.7	76.7	73.6	74.1	95.5	98.3	88.5	86.3
	SSDAN [38]	95.5	86.6	93.5	96.9	95.0	97.3	80.2	80.2	83.6	97.3	99.3	96.2	91.8
	ENT [55]	79.5	67.7	77.2	91.2	91.2	87.5	66.6	60.5	62.6	89.9	91.0	83.7	85.4
	CDAC [56]	85.4	80.6	75.2	83.8	90.0	88.5	65.8	65.5	72.7	86.1	80.6	92.0	79.8
	BSCA [14]	96.8	89.7	94.7	97.2	**97.2**	97.5	84.1	82.6	84.0	97.9	**99.7**	98.3	93.3
	ours	**97.1**	**93.7**	**97.1**	**97.4**	96.8	**98.0**	**85.2**	**82.6**	**87.0**	**99.0**	99.3	**99.0**	**94.3**

TABLE IV
RESULTS OF CFAN WITH EFFICIENTNET_B0 AS BACKBONE UNDER THREE-SHOT SETTING

Type	Method	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean
SSL	FixMatch [52]		72.2			93.2			63.0			96.9		81.3
UDA	ECB-FAM [53]	91.7	76.8	83.0	91.8	80.5	90.8	68.6	69.5	62.4	92.4	98.6	84.7	82.6
SSDA	S+T	91.1	81.5	86.0	92.6	84.2	91.4	71.4	71.9	66.2	93.1	97.6	90.4	84.8
	ADDA [54]	90.2	77.7	84.3	89.1	79.8	90.8	69.8	69.9	69.3	94.1	96.6	83.3	82.9
	RevGrad [38]	90.9	78.4	84.5	92.7	84.4	91.5	72.8	69.6	72.0	94.1	99.0	89.2	84.9
	SSDAN [38]	90.6	79.6	83.7	90.8	86.0	89.6	73.1	73.8	72.0	96.5	99.0	87.2	85.2
	ENT [55]	88.3	78.2	84.8	91.5	83.2	90.5	70.7	68.2	71.9	96.5	97.9	91.0	84.4
	CDAC [56]	87.0	84.7	86.0	88.5	78.6	84.8	68.5	66.3	68.5	88.9	92.7	81.2	81.3
	BSCA [14]	96.1	88.3	93.8	**96.5**	**95.8**	**96.4**	77.0	78.9	83.0	98.3	98.6	98.6	91.8
	ours	**96.6**	**94.2**	**94.1**	95.7	95.4	96.0	**82.7**	**79.9**	**83.3**	**98.6**	**99.3**	**99.0**	**92.9**

is a recently proposed method for remote sensing scene classification under the unsupervised cross-domain setting, which combines error-correcting boundaries with a feature adaptation metric, and outperforms other unsupervised scene classification methods. **S + T** is simply trained, with supervision from source domain and labeled target domain samples. **ADDA** [54] is the first method for semisupervised remote sensing DA. **RevGrad** [61] is a commonly used baseline in semisupervised remote sensing DA, which adopts adversarial training. Developed from MME [11], **SSDAN** [38] alternately maximizes the entropy of the classifier and minimizes that of the feature extractor. An early SSL method, based on the low-density assumption, **ENT** [55] improves classification accuracy by minimizing the entropy of unlabeled target data. **CDAC** [56] achieves SOTA performance in SSDA, minimizing the inter- and intradomain discrepancy through adversarial adaptive clustering loss. We believe that **BSCA** [14] is the current SOTA method for semisupervised remote sensing DA. Supervised and unsupervised modules align source and target domain distributions from different levels.

2) Results: Tables II–IV show comparison results of different backbones. CFAN performs best on three backbones, with a mean accuracy of 92.9%, 93.3%, and 94.3% on EfficientNet_B0, VGG16, and ResNet34, respectively.

a) Comparison results of different types of methods: SSL shows inferior performance compared with the SSDA methods, and is easily affected by the discrepancy between the two domains. The UDA method cannot perform stably in all

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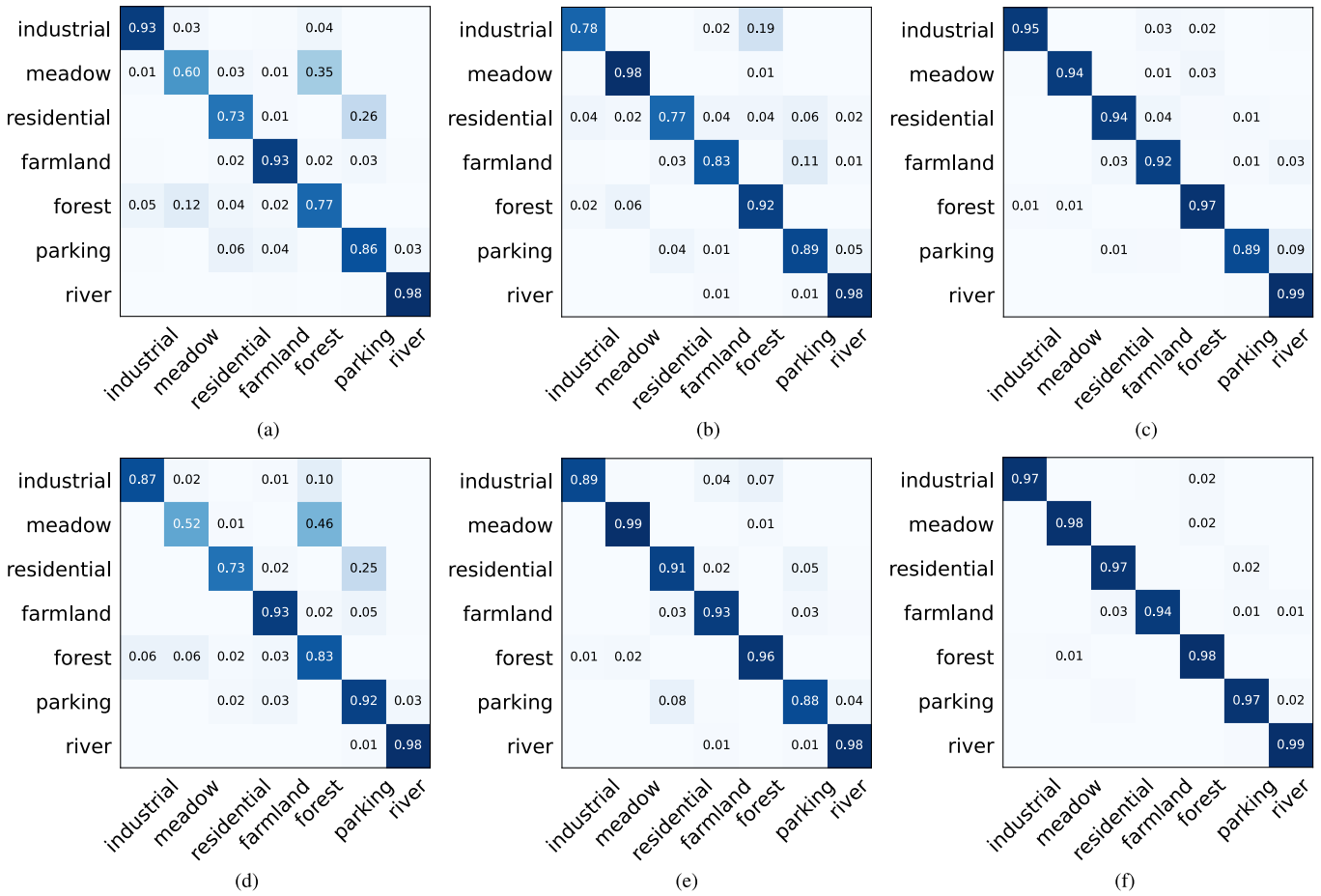

Fig. 4. Confusion matrices of CFAN and BSCA with ResNet34 on three adaptation scenarios. (a) BSCA ($A \rightarrow R$). (b) BSCA ($R \rightarrow N$). (c) BSCA ($W \rightarrow N$). (d) CFAN ($A \rightarrow R$). (e) CFAN ($R \rightarrow N$). (f) CFAN ($W \rightarrow N$).

scenarios. For instance, in Table III, the classification accuracy of adaptation scenario $W \rightarrow R$ is 65.4%, but $A \rightarrow W$ has an accuracy of 97.6%, which indicates that the UDA method cannot perform robustly. SSDA methods achieve more robust performance than UDA. Accuracies of all adaptation scenarios are relatively close, which demonstrates that annotating several target samples improves the model's generalization ability.

b) Comparison results of different backbones: From Tables II–IV, we can observe that CFAN performs best when taking ResNet34 as the backbone; the accuracy is 1% higher than VGG16 and 1.4% higher than EfficientNet_B0, because ResNet34 integrates multiple residual blocks that are easy to optimize, and connects residual structures with a shortcut operation. Hence, ResNet34 with residual blocks can significantly improve the ability of feature extraction.

c) Comparison results with BSCA: CFAN outperforms BSCA on three backbones, with the best accuracy, 94.3%, on ResNet34. As seen in Table III, CFAN outperforms BSCA by 1%–3% in most adaptation scenarios. Although our method shows inferior performance compared with BSCA in several tasks, the gaps are limited within 0.4%, which indicates that CFAN achieves comparable performance. When RSSCN7 is taken as the target domain, the classification accuracy is lower than in other adaptation scenarios because the images of RSSCN7 show larger variances in color, shape, and texture. But CFAN still outperforms BSCA, especially in the task of $W \rightarrow R$. Fig. 4 shows the confusion matrices of CFAN and BSCA with ResNet34 on adaptation scenarios $A \rightarrow R$, $R \rightarrow N$, and $W \rightarrow N$. Note that CFAN can obtain clearer confusion matrices than BSCA, demonstrating that CFAN decreases the discrepancy between the source and target domains by learning better domain-invariant representations.

3) Visualization of Features via t-SNE: Category-level alignment is critical to our method. Therefore, we use t-SNE [62] to intuitively show the feature distributions of different methods with ResNet34 on adaptation scenarios $A \rightarrow R$, $R \rightarrow N$, $W \rightarrow N$, and $W \rightarrow R$ in Fig. 5.

S + T shows that the features of different classes are not clustered well, and classification boundaries are also indistinguishable. For example, some Industrial samples from the target domain are mixed with the Residential class in scenario $A \rightarrow R$, but with the Meadow class in scenario $W \rightarrow N$. Compared with S + T, BSCA obtains a better effect of feature clustering, which increases the interclass variance. However, the classification boundary is not clear enough, especially in adaptation scenarios $R \rightarrow N$ and $W \rightarrow R$. Besides, many class cluster centers are close to each other. Although some features are still wrongly classified, CFAN achieves the best adaptation performance, with much clearer classification boundaries.

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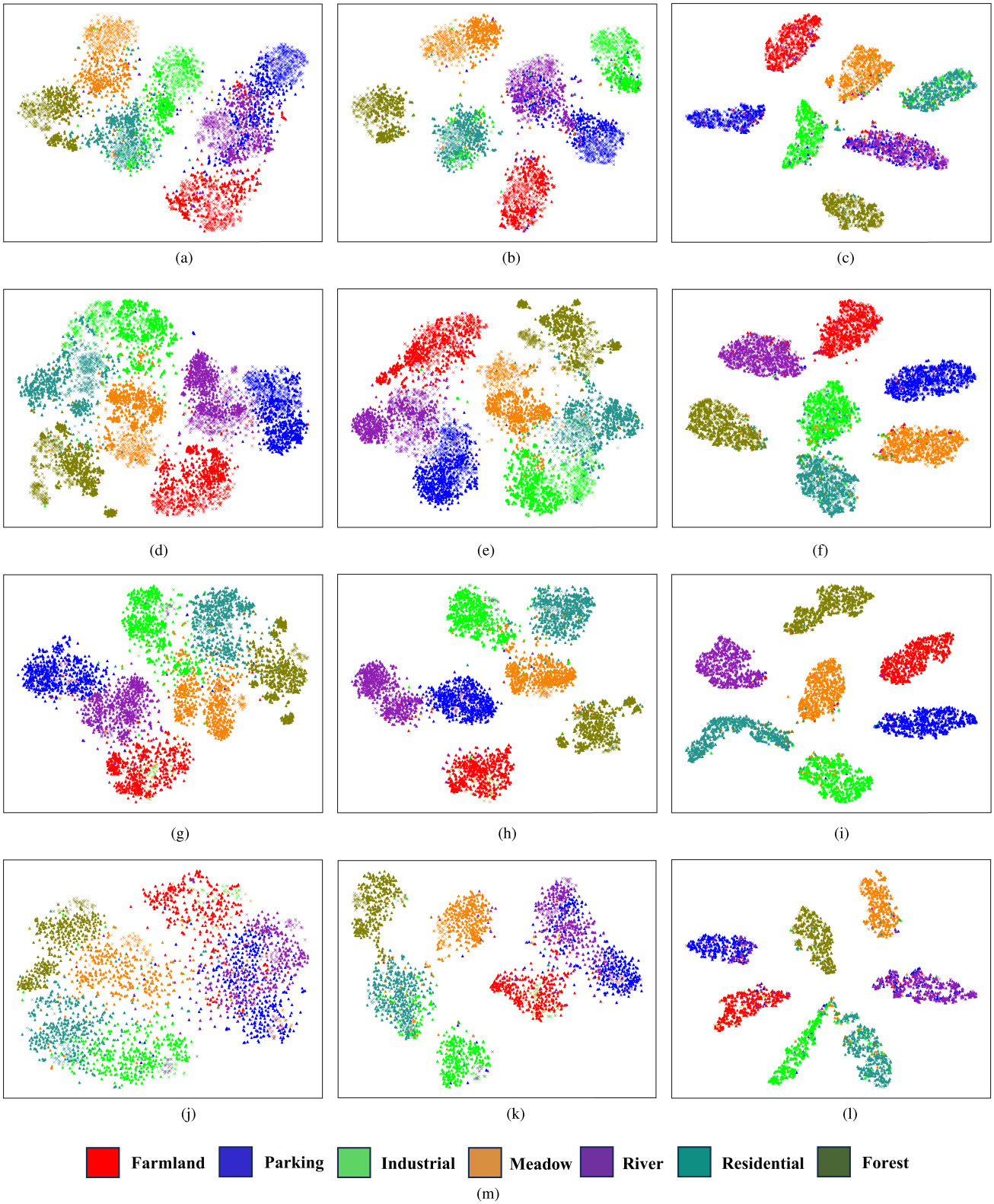

Fig. 5. Visualization of features of different methods on four adaptation scenarios. × and ▲ denote features extracted from source and target domain samples, respectively. (a) S + T ($A \rightarrow R$). (b) BSCA ($A \rightarrow R$). (c) CFAN ($A \rightarrow R$). (d) S + T ($R \rightarrow N$). (e) BSCA ($R \rightarrow N$). (f) CFAN ($R \rightarrow N$). (g) S + T ($W \rightarrow N$). (h) BSCA ($W \rightarrow N$). (i) CFAN ($W \rightarrow N$). (j) S + T ($W \rightarrow R$). (k) BSCA ($W \rightarrow R$). (l) CFAN ($W \rightarrow R$).

D. Ablation Study

To evaluate the impact of each component on total loss, we conducted a series of ablation studies, taking ResNet34 as the backbone. The results are shown in Table V, where ✓ means the corresponding loss is applied.

Based on the labeled source and target domain samples, we only used $\mathcal{L}_{ce}$ to train the neural network, and model 1 obtained a mean accuracy of 88.0% over all adaptation scenarios. This demonstrates that the generalization ability of model 1 is limited, since only a few labeled target samples

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TABLE V
ABLATION STUDY OF CFAN WITH RESNET34 AS BACKBONE UNDER THREE-SHOT SETTING

Model	L_{ce}	L_{pl}	L_{alg}	L_{ad}	$A \rightarrow N$	$R \rightarrow N$	$W \rightarrow N$	$N \rightarrow A$	$R \rightarrow A$	$W \rightarrow A$	$N \rightarrow R$	$A \rightarrow R$	$W \rightarrow R$	$N \rightarrow W$	$A \rightarrow W$	$R \rightarrow W$	Mean
1	✓				92.0	81.5	88.5	93.8	86.1	95.1	76.3	76.5	76.5	96.5	99.7	93.4	88.0
2	✓	✓			95.4	92.0	94.1	97.2	96.4	97.7	83.4	82.9	83.2	98.3	99.0	98.3	93.2
3	✓	✓	✓		96.1	91.8	96.1	97.3	95.7	97.4	84.3	81.5	84.9	98.6	99.3	97.6	93.3
4	✓	✓		✓	96.5	**94.1**	97.1	97.3	**96.9**	97.7	**86.0**	**83.2**	84.9	99.0	99.3	97.6	93.7
5	✓	✓	✓	✓	**97.1**	93.7	**97.1**	**97.4**	96.8	**98.0**	85.2	82.6	**87.0**	**99.0**	**99.3**	**99.0**	**94.3**

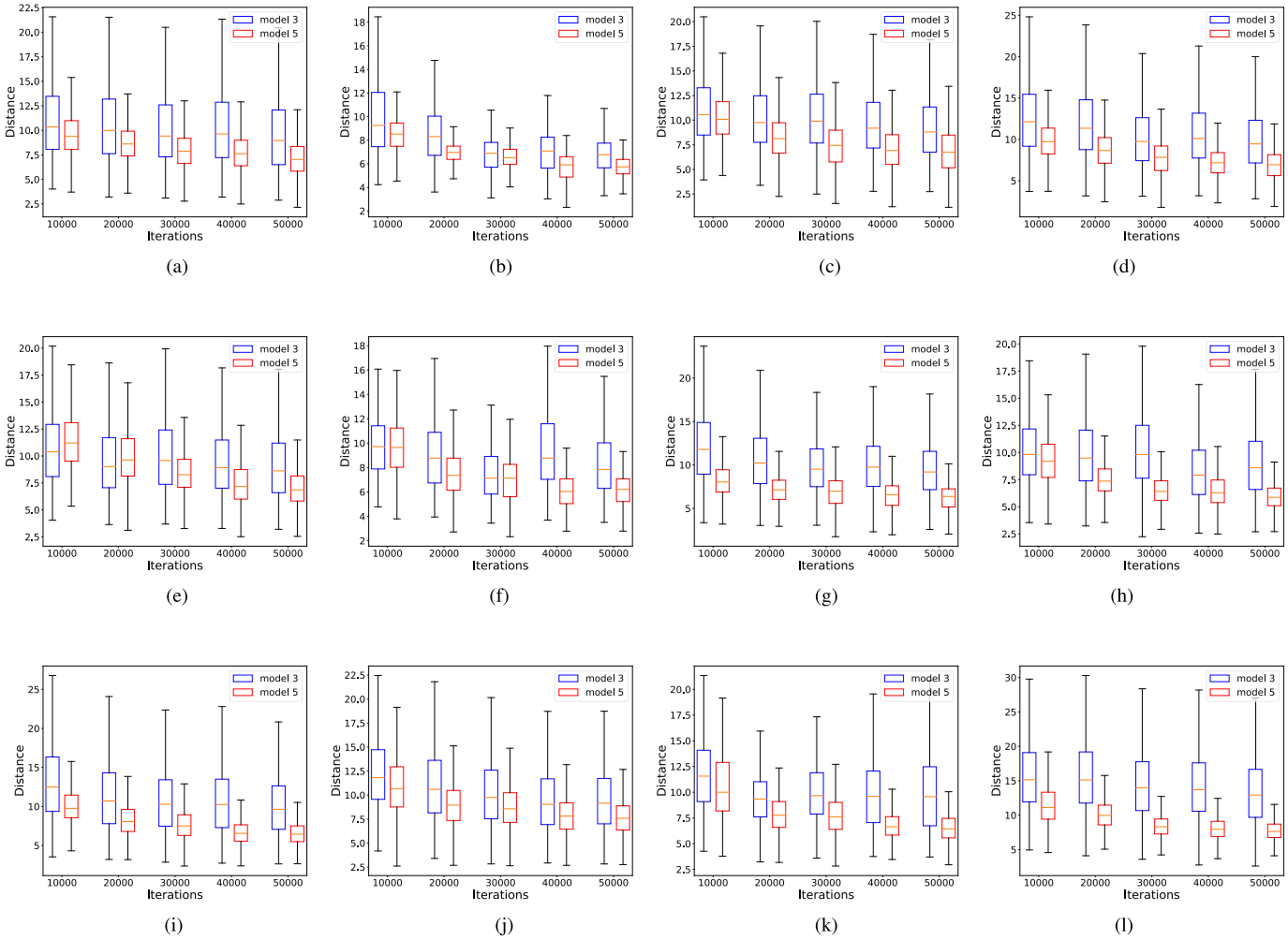

Fig. 6. Visualization of quality of target prototypes. Blue rectangles: model 3; red rectangles: model 5. (a) $A \rightarrow N$. (b) $A \rightarrow W$. (c) $A \rightarrow R$. (d) $N \rightarrow A$. (e) $N \rightarrow R$. (f) $N \rightarrow W$. (g) $W \rightarrow A$. (h) $W \rightarrow R$. (i) $W \rightarrow N$. (j) $R \rightarrow A$. (k) $R \rightarrow W$. (l) $R \rightarrow N$.

are applied. Then we added $\mathcal{L}_{\text{pl}}$ to model 1, which makes full use of the unlabeled target domain samples. $\mathcal{L}_{\text{pl}}$ reduces the entropy of pseudo-labeled samples, and drives the classification boundary through the low-density sample space. The mean accuracy of model 2 is significantly increased by 5.2%, indicating that the generalization ability of model 2 is greatly improved. We also find that both $\mathcal{L}_{\text{alg}}$ and $\mathcal{L}_{\text{ad}}$ improve the model's performance to varying degrees. Note that model 3 achieves a slight performance gain of 0.1% after $\mathcal{L}_{\text{alg}}$ is embedded, perhaps because of the low-quality pseudo labels generated by $\mathcal{L}_{\text{pl}}$, which are not conducive to building accurate and robust category prototypes for alignment. Moreover, by combining $\mathcal{L}_{\text{ad}}$ with $\mathcal{L}_{\text{ce}} + \mathcal{L}_{\text{pl}}$, model 4 surpasses model 2 by 0.5%. This demonstrates that $\mathcal{L}_{\text{ad}}$ can effectively achieve domain alignment, and align multimodal structures with different distributions. Finally, when the overall loss is used to train the neural network, model 5 achieves the best performance, which is 6.3% higher than model 1, and 1.1% higher than model 2. This was because the adversarial training module first performs domain alignment and aligns the multimodal structures of different domains. This enables the generation of higher quality pseudo labels, so as to build accurate target prototypes, further promoting category-level feature alignment.

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TABLE VI
QUALITY OF TARGET PROTOTYPES OF BSCA AND CFAN

Method	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$
BSCA	22.2	22.0	21.5	22.0	22.0	21.5	21.2	20.2	19.5	20.6	21.3	21.8
CFAN	7.2	7.8	6.6	6.9	7.7	6.2	7.1	6.9	6.0	6.2	5.8	6.6

TABLE VII
ABLATION STUDY OF CFAN WITH SWIN TRANSFORMER AS BACKBONE UNDER THREE-SHOT SETTING

Backbone	L_{ce}	L_{pl}	L_{alg}	L_{ad}	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean
Swin Transformer	✓				78.3	69.4	65.9	75.5	68.4	71.0	64.7	64.5	52.7	81.3	94.8	75.3	71.8
	✓	✓			83.3	72.6	67.6	78.5	74.1	74.7	65.0	69.4	55.7	78.5	95.8	76.7	74.3
	✓	✓	✓		83.0	73.7	68.4	78.3	73.8	76.7	63.8	70.7	56.3	81.3	96.9	77.1	75.0
	✓	✓	✓	✓	82.8	74.1	74.0	79.0	75.1	76.6	68.6	71.8	59.4	80.2	96.5	75.3	76.1

TABLE VIII
RESULTS OF THRESHOLD ANALYSIS EXPERIMENTS WITH RESNET34 AS BACKBONE UNDER THREE-SHOT SETTING

Threshold τ	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean
0.3	96.8	91.5	96.1	**97.5**	96.5	97.5	85.0	83.8	86.0	98.6	99.3	97.2	93.8
0.4	96.3	92.9	96.5	97.0	96.2	97.6	84.3	83.0	86.8	98.6	99.3	97.6	93.8
0.5	**97.1**	93.7	**97.1**	97.4	**96.8**	**98.0**	**85.2**	82.6	**87.0**	**99.0**	**99.3**	**99.0**	**94.3**
0.6	95.5	93.4	94.9	96.9	96.3	97.7	85.0	**84.4**	84.9	97.9	99.3	97.9	93.7
0.7	96.1	**93.8**	96.0	95.9	96.0	96.5	81.8	81.3	82.7	97.6	99.3	97.2	92.8

E. Target Prototypes Analysis

We take the L_2 distance between target domain samples and target prototypes to measure the quality of target prototypes. The visualization results are shown in Fig. 6. The smaller the distance, the more accurate the target prototypes. We show the boxplots of models 3 and 5 over 12 adaptation scenarios in Fig. 6. Comparing these boxplots, it is clear that model 5 has a tighter data distribution than model 3, and the overall data is smaller. The smaller median of model 5 suggests that the target domain prototypes it generates are closer to the target domain samples, and thus are more accurate. As training progresses, the data of model 5 decreases more significantly than those of model 3. We also compare the quality of target domain prototypes between BSCA and CFAN to demonstrate that CFAN improves the quality of pseudo labels. We compute the mean L_2 distance between target domain samples and target prototypes. Experiment results are presented in Table VI. The mean L_2 distance of CFAN is significantly smaller than BSCA in all adaptation scenarios. This indicates that the inclusion of the adversarial training module reduces the noise in pseudo labels, and further improves the quality of the target domain prototypes.

F. Transformer-Based Backbone

We explore the effect of a transformer-based backbone in our setting, taking a Swin Transformer to test the proposed CFAN and conduct ablation studies to analyze different submodules. The experimental results are shown in Table VII. We only applied L_{ce} in the model, which achieved 71.8% accuracy. When L_{pl}, L_{alg}, and L_{ad} were employed sequentially, model performance gradually improved to 74.3%, 75.0%, and 76.1%, respectively. From the ablation study of the Swin Transformer, we can draw the same conclusion as for the CNN-based backbone. Nevertheless, compared to other CNN-based backbones, CFAN exhibited inferior performance overall, possibly due to the limited amount of training data, which hinders the full activation of the transformer [63]. Transformer-based models lack image-specific inductive bias [64], and the transferability of the adversarial feature space of vision transformer models is being investigated [65].

G. Parameter Sensitivity Study

We conducted a series of experiments to analyze the effect of different entropy thresholds τ, which is used in 8. We set the entropy threshold τ to 0.3, 0.4, 0.5, 0.6, and 0.7. The experimental results are shown in Table VIII. We can find that both low and high entropy thresholds obtain inferior performance. A low entropy threshold can ensure the quality of pseudo labels, but it leads to imbalanced learning of different classes, and limits utilization of pseudo labels, especially in the early training stage. A high entropy threshold inevitably introduces more wrong pseudo labels, confusing the training process of the neural network. Therefore, we select the entropy threshold as 0.5 by default, which obtains the best mean accuracy over all adaptation scenarios.

H. Running Time Tests

We conducted experiments to compare the running time of S + T, BSCA, and CFAN, testing 12 adaptation scenarios for each method. S + T and BSCA were trained for 10 000 steps, and CFAN for 50 000 steps. We computed the mean running time of 12 adaptation scenarios. Experimental results

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TABLE IX
RUNNING TIME OF S + T, BSCA, AND CFAN

Method	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean Accuracy	Mean Running Time (h)
S+T	91.2	79.8	85.8	92.9	84.5	94.8	74.0	75.0	75.0	95.1	98.3	90.3	86.4	1.9
BSCA	96.6	91.0	94.4	97.5	96.8	97.9	85.1	83.3	85.3	99.0	98.6	97.9	93.6	2.5
CFAN (10000 iterations)	95.7	93.9	95.5	97.0	96.3	97.1	83.2	82.0	84.1	98.6	99.0	96.9	93.3	2.5
CFAN (30000 iterations)	96.5	94.2	95.4	97.4	95.9	97.4	84.9	83.2	85.9	98.3	99.0	97.9	93.8	7.2
CFAN (50000 iterations)	97.1	93.7	97.1	97.4	96.8	98.0	85.2	82.6	87.0	99.0	99.3	99.0	94.3	12.1

TABLE X
COMPARATIVE RESULTS OF DIFFERENT NUMBERS OF LABELED TARGET DOMAIN SAMPLE WITH RESNET34 AS BACKBONE (BSCA/OURS)

Num.	$A \to N$	$R \to N$	$W \to N$	$N \to A$	$R \to A$	$W \to A$	$N \to R$	$A \to R$	$W \to R$	$N \to W$	$A \to W$	$R \to W$	Mean
0	95.4/96.7	82.1/88.1	93.6/96.5	96.2/97.2	93.7/96.2	97.4/97.6	79.0/78.9	73.8/75.9	81.1/84.9	99.0/97.2	99.0/99.7	95.8/96.5	90.5/92.1
1	96.2/96.5	87.4/90.0	93.3/97.0	96.4/97.4	94.6/95.9	97.5/97.4	82.6/81.7	80.5/81.5	83.5/83.6	98.6/99.0	98.6/**99.7**	95.8/99.0	92.1/93.2
3	96.8/**97.1**	89.7/93.7	94.7/97.1	97.2/97.4	97.2/**96.8**	97.5/98.0	84.1/85.2	82.6/82.6	84.0/87.0	97.9/99.0	99.7/99.3	98.3/**99.0**	93.3/94.3
5	95.6/96.8	90.8/94.5	94.4/96.5	98.1/98.0	96.1/96.7	97.5/98.8	84.4/83.7	82.8/83.1	84.5/86.9	97.5/97.2	99.0/99.0	98.2/98.3	93.2/94.1
10	95.7/96.9	92.3/**95.0**	96.1/**97.2**	97.1/**98.5**	97.2/96.3	97.8/**98.9**	84.8/**86.0**	83.2/**84.5**	85.3/**88.8**	100.0/**99.2**	99.2/99.2	98.3/97.9	93.9/**94.9**

are presented in Table IX. The mean running times for S + T, BSCA, and CFAN were 1.9, 2.5, and 12.1 h, respectively. Although the training time of CFAN was much longer than the others, all methods had the same testing time, since there was no gradient computation or parameter updating during testing. Furthermore, the mean running times per step for S + T, BSCA, and CFAN were 0.7, 0.9, and 0.9 s, respectively, which are roughly consistent.

We also ran CFAN for 10 000 and 30 000 iterations, which achieved the respective mean accuracies of 93.3%–93.8%. Although the mean running times were reduced to 2.5 and 7.2 h, respectively, the mean accuracy of 12 adaptation scenarios decreased.

I. Number of Labeled Target Samples

Table X provides experimental results with 1, 3, 5, and 10 labeled target samples, where the mean accuracy improved from 93.2% to 94.9%. CFAN and BSCA performed worst when we annotated only 1 target domain image. The accuracy of most adaptation scenarios improved when the number of annotations reached 10. Especially in the adaptation scenario of $* \to R$, as the amount of labeled target domain data increased, performance improved significantly, and CFAN achieved larger improvements than BSCA. Specifically, BSCA only improved by 1.8%–2.7%, while CFAN improved by 2.8%–5.2%.

We also conducted experiments to compare BSCA and CFAN in an unsupervised setting by setting the number of labeled target domain sample to 0. CFAN obtained a mean accuracy of 92.1%, which was 1.1% less than with 1 labeled target domain sample. BSCA achieved a mean accuracy of 90.5%, indicating that both methods can be adapted to an unsupervised setting with only a slight decrease in performance. CFAN outperformed BSCA in most scenarios, especially $R \to N$. This demonstrates that CFAN is more effective than BSCA in both unsupervised and semisupervised settings.

V. CONCLUSION

We presented a semisupervised cross-domain remote sensing scene classification method, CFAN, aiming to achieve category-level feature alignment and design a dual-directional prototype alignment module. The pseudo-labeling module fully utilizes unlabeled target domain samples and generates pseudo labels, which are used to create prototypes in the target domain. We align the multimodal structure between different domains to improve the quality of pseudo labels in the adversarial training module. In experiments conducted on four commonly used remote sensing datasets, the proposed method outperformed the current SSL, UDA, and SSDA methods. In the future work, we will seek to improve the training efficiency of CFAN, and will explore the transferability of the adversarial feature space of vision transformer models.

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